

Visipoint Performance Test — Bug Report

Test date:	2026-08-02
Test type:	Load test — 1000 concurrent virtual users, 30s ramp-up, single account, Login → Logout per user
Tool:	Apache JMeter 5.6.3
Test plan:	D:\apache-jmeter-5.6.3\visipoint-project\visipoint-load-test.jmx
Raw results:	results.jtl (contains duplicate-logging artifact, see Appendix) / results-clean.csv (deduplicated, used below)
Tester:	Claude (JMeter load test, built from a real captured login/logout request; user executed the run)

Summary Results

Step	Success	Failed	Avg	P50	P95	P99	Max
Login	864/1000 (86.4%)	136 (13.6%)	34.0s	22.9s	88.6s	152.5s	299.4s
Logout	119/1000 (11.9%)	881 (88.1%)	11.0s	0.7s	72.5s	77.7s	175.2s

BUG-PERF-001 — Concurrent logins from the same account invalidate each other's session, breaking Logout with 401

Module	Passport Account — Authentication / Session Management
Endpoint	GET https://api.visipoint.me/api/logout (session via POST https://api.visipoint.me/api/login)
Severity	High
Status	New

#	Step	Expected	Actual
1	Send 1000 concurrent POST /api/login requests for the same account , ramped over 30s	Each login succeeds, returns a valid, independently-usable access_token	864/1000 (86.4%) returned 200 OK with a token — server accepted the concurrent logins
2	Immediately after login, send GET /api/logout using that thread's own token	Each logout succeeds (200 OK), invalidating only that session	881/1000 (88.1%) failed. 869 returned a clean 401 Unauthorized — not a timeout or connection error

3	Compare failure pattern against timing	If tokens simply expired, failures would be random/timing-independent	Failure rate (88%) tracks the volume of concurrent logins on the same account — consistent with each new login invalidating the previous session's token
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Expected: Each of the 1000 simulated logins/sessions should be independent — logging in a second time (e.g., another device/tab) should not silently invalidate a different active session for the same account.

Actual: 88.1% of logout attempts fail with `401 Unauthorized`, indicating concurrent logins to the same account invalidate each other's tokens server-side.

Root cause (assessed, not yet confirmed against backend code): The auth/session backend appears to allow only one active token per account (or otherwise invalidates prior tokens on new login) rather than maintaining independent, concurrently-valid sessions. This is a **session-management architecture issue**, not a load-capacity issue — Login itself succeeded cleanly in the large majority of cases (86.4% with 200 OK, zero 401s on Login). The system correctly processes each login; as a side effect, it disqualifies the previous session.

Why this matters beyond the load test: This same mechanism would affect any real user logging into the same account from two places in quick succession (two browser tabs, phone + laptop) — the first session could unexpectedly fail with 401 on its next authenticated call once the second login completes. Recommend verifying this against normal (non-load-test) multi-session usage — it may reproduce with as few as 2 concurrent logins, not just 1000.

BUG-PERF-002 — Login response time degrades severely under concurrent load (median 22.9s, P99 152.5s, max ~5 min)

Module	Passport Account — Authentication
Endpoint	POST <code>https://api.visipoint.me/api/login</code>
Severity	Medium-High (see attribution caveat below)
Status	New

#	Step	Expected	Actual
1	Ramp up 1000 concurrent login requests over 30 seconds	Response times stay reasonably close to baseline, or degrade gracefully	Median (P50): 22.9s . P95: 88.6s. P99: 152.5s. Max: 299.4s (~5 min) for a single login
2	Track throughput (completed)	Throughput stays roughly stable or degrades gracefully	Throughput collapsed : ~26 req/s in the first ~30s window, down to 0.5 req/s in the final ~3.5-minute window

	requests/sec) over the run		
3	Check whether failures were server rejections or connection-level	Explicit server rejection under load would show as HTTP error codes (429, 503, etc.)	136/1000 (13.6%) Login failures were connection-level (HttpException, SocketException) — no HTTP error codes; the server never explicitly refused a login

Expected: A login request should complete in roughly 1-2 seconds under normal conditions; under 1000-concurrent-user load, some degradation is expected but should not extend into tens of seconds to minutes for a majority of requests.

Actual: Response times ballooned by 1-2 orders of magnitude as load increased, with throughput visibly collapsing rather than stabilizing — consistent with request queuing/backlog building faster than the system could drain it.

Root cause — NOT fully isolated, attribution caveat: Two possible contributing causes cannot be conclusively separated from this single-machine test:

1. **Backend capacity limit** — the Visipoint auth backend genuinely struggling with 1000 concurrent logins (DB connection pool exhaustion, per-login encryption/decryption CPU cost, etc.)
2. **Load-generator capacity limit** — a single machine originating 1000 concurrent HTTPS connections is itself resource-intensive (TLS handshake overhead, OS socket/file-descriptor limits, JMeter's default connection pool sizing)

To resolve the ambiguity: re-run with either (a) a distributed load generator (multiple machines) to rule out client-side bottlenecks, or (b) server-side monitoring (CPU, DB connection pool usage, request queue depth) captured during the run to directly observe where time is being spent.

Appendix: JMX listener duplication artifact (test-tooling issue, not a product bug). The raw `results.jtl` file contains exactly 2x the real sample count (4000 rows instead of 2000) because the test plan had two listeners both writing to a file named `results.jtl` — one embedded in the JMX ("View Results Tree"), one from the `-l results.jtl` CLI flag. The two listeners used slightly different column layouts, so interleaved rows in the raw file are misaligned every other row. All analysis in this report used a filtered, deduplicated dataset (`results-clean.csv` , 1000 Login + 1000 Logout rows, verified against the live console summary counts). Future test plans should use only one results listener.